Uncovering Banking Fraud Patterns: An Exploratory Data Analysis

A Python based Project

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1. Abstract

The increasing use of digital banking and electronic transactions has made transaction monitoring and fraud-risk analysis important areas of financial data analytics. This project, titled *Uncovering Banking Fraud Patterns: An Exploratory Data Analysis*, focuses on analysing banking transaction data to identify patterns and relationships associated with fraudulent and non-fraudulent transactions.

The project uses a dataset containing 50,000 transaction records and 21 original features, including transaction amounts, account balances, failed transaction counts, card age, risk scores, transaction types, locations and fraud labels. Python libraries, including Pandas, Matplotlib, Seaborn and Plotly, were used to prepare, analyse and visualize the data.

The analysis involved data cleaning and validation, descriptive statistical analysis, category-based fraud-rate comparisons, monthly transaction trend analysis and correlation analysis. The visualizations revealed that 32.13% of the transactions were labelled as fraudulent. Fraud-labelled transactions generally exhibited higher risk scores and higher failed-transaction counts. The correlation analysis identified a positive correlation of approximately 0.51 between failed transaction count over the previous seven days and the fraud label, followed by a correlation of approximately 0.39 for risk score. Fraud rates were relatively similar across locations, transaction types and card-age groups.

The findings provide an exploratory understanding of transaction characteristics associated with fraud labels and demonstrate how Python-based data analytics can support the investigation of potential fraud indicators. The results describe patterns within the dataset and do not establish causation or represent the performance of a validated predictive model.

2. Introduction

Banking transactions generate substantial amounts of data containing information about transaction values, customer accounts, payment activities, authentication methods and other security-related characteristics. Analysing this data can help financial institutions understand transaction behaviour, identify unusual patterns and investigate potential indicators of fraudulent activity.

Banking fraud analytics involves examining transaction data to understand the characteristics and patterns associated with fraudulent transactions. Variables such as transaction amount, failed transaction count, account balance and risk score may provide useful information for investigating differences between fraudulent and non-fraudulent transactions. However, the usefulness of individual variables depends on the characteristics of the dataset and the way the variables have been defined or generated.

Exploratory Data Analysis (EDA) is an important stage of the data analytics process. It involves inspecting datasets, calculating descriptive statistics, examining distributions, comparing categories and investigating relationships between variables. Data visualization complements this process by presenting numerical information through charts and graphs, making patterns and differences easier to interpret.

This project, *Uncovering Banking Fraud Patterns: An Exploratory Data Analysis*, uses Python to examine a banking fraud detection and risk analytics dataset. The project begins with data cleaning and validation, followed by statistical analysis and visual exploration of transaction characteristics. It investigates fraud-label distributions, transaction amount patterns, risk-score distributions, failed transaction counts, account balances, fraud rates across categories, monthly transaction volumes and correlations between numerical variables.

The project aims to develop a clearer understanding of the dataset and identify variables and categories that show noticeable associations with the fraud label. The findings are intended to demonstrate the application of exploratory data analysis to a financial analytics problem and provide a foundation for further investigation and future predictive modelling.

3. Project Statement

Banking transaction datasets contain multiple variables that may help explain differences between fraudulent and non-fraudulent transactions. However, these variables must be examined systematically to understand their distributions, identify unusual values, compare transaction categories and investigate relationships with the fraud label.

The central problem addressed by this project is the need to explore and interpret banking transaction data to identify patterns and characteristics associated with transactions labelled as fraudulent. Without appropriate data preparation and exploratory analysis, it may be difficult to understand the distribution of fraud labels, evaluate differences between transaction groups or identify variables that warrant further investigation.

Therefore, this project applies data cleaning, descriptive statistical analysis, category-based comparisons, correlation analysis and data visualization to a banking fraud detection dataset. The analysis focuses on understanding transaction behaviour and examining how variables such as risk score, failed transaction count, transaction amount, account balance and card age relate to the fraud label.

The project is limited to exploratory analysis of the available dataset. It does not aim to establish causation, verify actual fraudulent activity independently or develop a validated fraud prediction model. Instead, it seeks to identify meaningful patterns within the data and communicate the findings in a clear and structured manner.

4. Project Objectives

The primary objective of this project is to explore banking transaction data using Python to identify patterns and relationships associated with fraudulent and non-fraudulent transactions.

The specific objectives are:

1. **To clean and prepare the dataset:** Inspect and validate data types, missing values, duplicate records, categorical values, binary variables and numerical ranges to improve data quality.
2. **To perform Exploratory Data Analysis:** Examine the structure and characteristics of the dataset using descriptive statistics, frequency distributions and grouped calculations.
3. **To analyse fraud-label distribution:** Determine the number and percentage of fraudulent and non-fraudulent transactions in the dataset.
4. **To investigate transaction characteristics:** Examine transaction amounts, account balances, failed transaction counts, card age and risk scores, including comparisons between fraud-label groups.
5. **To compare fraud rates across categories:** Investigate differences in fraud rates across locations, transaction types, card-age groups and risk-score categories.
6. **To examine transaction trends over time:** Analyse monthly transaction volumes throughout the year 2023 using the timestamp information available in the dataset.
7. **To investigate relationships between variables:** Use correlation analysis to examine linear relationships between numerical variables and the fraud label.
8. **To visualize analytical findings:** Develop charts and graphs using Matplotlib, Seaborn and Plotly to communicate distributions, comparisons, relationships and trends.
9. **To derive data-driven insights:** Summarize the findings, identify variables that may warrant further investigation and discuss the potential relevance of the results to banking fraud-risk analysis.
10. **To recognize analytical limitations:** Distinguish observed associations from causal relationships and identify considerations for further validation and future predictive modelling.

5. Dataset Description

5.1 Dataset Source

The dataset used in this project is the Fraud Detection Transactions Dataset, obtained from Kaggle. It contains banking transaction records and related attributes that can be explored to understand differences between fraudulent and non-fraudulent transactions.

The dataset includes transaction details, account activity, transaction context, risk-related information and a fraud label. These attributes provide the basis for examining transaction distributions, comparing fraud rates across categories, investigating relationships between variables and identifying patterns that may be relevant to fraud-risk analysis.

The dataset is used for exploratory data analysis and visualization with Python. The analysis focuses on the information available in the dataset and does not assume that every transaction labelled as fraudulent has been independently verified.

5.2 Dataset Dimensions

The original dataset contains 50,000 records and 21 columns, representing transaction-level observations and their associated attributes. Each record represents a transaction, while the columns describe transaction values, account activity, transaction characteristics and fraud-related indicators.

The dataset dimensions are summarized below;

Attribute	**Description**
Total Rows	50000
Total Columns	21
Unit of Observation	Individual transaction record
Dataset Type	Structured banking transaction data
Target Variable	Fraud_Label
Target Variable Values	0 = Non-fraudulent, 1 = Fraudulent
Data Source	Kaggle

The original dataset dimensions are \(50{,}000 \times 21\). Additional columns, including Card_Age_Group, Risk_Category and Month, were created during the analysis for grouping and visualization purposes. These derived columns are not part of the original 21 features.

5.3 Description of Important Columns

The dataset contains numerical, categorical, date-time and binary variables. The following table describes the original columns most relevant to the project.

Column Name	Data Type / Category	Description
Transaction_ID	Identifier	Unique identifier associated with a transaction.
User_ID	Identifier	Identifies the user associated with the transaction.
Timestamp	Date-time	Records the date and time when the transaction occurred.
Transaction_Amount	Numerical	Amount of money involved in the transaction.
Account_Balance	Numerical	Account balance associated with the transaction record.
Daily_Transaction_Count	Numerical	Number of transactions recorded for the day.
Avg_Transaction_Amount_7d	Numerical	Average transaction amount over the preceding seven days.
Failed_Transaction_Count_7d	Numerical	Number of failed transactions recorded over the preceding seven days.
Card_Age	Numerical	Age of the card, measured in days.
Transaction_Distance	Numerical	Distance-related measure associated with the transaction.
Risk_Score	Numerical	Risk score associated with the transaction.
Transaction_Type	Categorical	Type of transaction, such as ATM withdrawal, bank transfer, online or POS.
Device_Type	Categorical	Type of device associated with the transaction.
Location	Categorical	Location recorded for the transaction.

Merchant_Category	Categorical	Category of the merchant associated with the transaction
Card_Type	Categorical	Type or network category of the payment card, as recorded in the dataset.
Authentication_Method	Categorical	Authentication method used for the transaction
IP_Address_Flag	Binary	Indicates the IP-address-related flag recorded for the transaction.
Previous_Fraudulent_Activity	Binary	Indicates whether previous fraudulent activity was recorded.
Is_Weekend	Binary	Indicates whether the transaction occurred on a weekend.
Fraud_Label	Binary (target)	Indicates the transaction's assigned fraud label: 0 for non-fraudulent and 1 for fraudulent.

The dataset provides a combination of transaction-level numerical measures, categorical attributes and fraud-related indicators. The Fraud_Label column serves as the target variable for the exploratory comparisons in this project. The remaining variables are examined to understand their distributions and associations with the fraud label.

The dataset structure provides the foundation for the subsequent stages of data cleaning, validation, exploratory data analysis and visualization.

6. Tools and Technologies Used

The project used Python and its data analysis and visualization libraries to clean, explore and analyse the banking transaction dataset. The following tools and technologies supported the data preparation, statistical analysis and visualization stages.

6.1 Google Colab

Google Colab is a cloud-based environment that allows users to write and execute Python code in an interactive notebook. It was used to perform data cleaning, exploratory data analysis, calculations and visualization without requiring a local Python setup.

6.2 Python

Python is a high-level programming language widely used for data analysis and automation. In this project, it was used to process the dataset, perform analytical calculations and generate visualizations.

6.3 Pandas

Pandas is a Python library used for data manipulation and analysis through structures such as DataFrames. It was used to load, clean and validate the dataset, calculate summary statistics, group transactions by categories and prepare data for visualization.

6.4 NumPy

NumPy is a Python library that supports numerical calculations and array-based operations. It was available as part of the analytical environment and supports numerical data processing in Python; its use in specific project calculations should be documented according to the operations performed in the notebook.

6.5 Matplotlib

Matplotlib is a Python visualization library used to create charts and graphs. It was used to present transaction distributions and comparisons in a visual format, helping make patterns in the dataset easier to interpret.

6.6 Seaborn

Seaborn is a statistical data visualization library built on Matplotlib. It was used to create visualizations such as countplots, histograms, density plots, boxplots, violin plots, scatterplots and correlation heatmaps.

6.7 Plotly

Plotly is a visualization library that supports interactive charts. It was used to create bar and line charts with percentage labels and hover information, allowing transaction counts and fraud rates to be explored more conveniently.

7. Data Cleaning and Preprocessing

Data cleaning and preprocessing were carried out before exploratory data analysis (EDA) to improve the consistency, completeness, and reliability of the dataset. The process was performed in Python using the Pandas and NumPy libraries in Google Colab. The main activities included inspecting the dataset, identifying and handling missing values, checking for duplicate records, validating data types and values, investigating potential outliers, and performing final quality checks.

7.1 Initial Dataset Inspection

The dataset was loaded into a Pandas DataFrame from a CSV file in Google Drive. Initial inspection was performed using functions such as head(), shape, columns, and info() to understand the structure of the data, review sample records, identify the available variables, and examine their data types and completeness.

The initial dataset contained **50,000 rows and 21 columns**. It included transaction identifiers, transaction amounts and types, timestamps, account balances, device and location information, transaction activity indicators, risk scores, and the fraud label. This inspection established the dataset's initial structure and helped identify areas requiring further validation.

7.2 Identification and Treatment of Missing Values

Missing values were identified using Pandas functions such as isnull().sum() and isnull().mean(). The initial inspection identified missing values in four columns: Transaction_Amount, Device_Type, Daily_Transaction_Count, and Card_Type.

The missing values were handled according to the type of variable:

- **Transaction amount:** The 81 missing values were replaced with the median transaction amount of 69.61. The median was selected as a measure of central tendency that is less sensitive to unusually large transaction values than the mean.
- **Daily transaction count:** The 30 missing values were replaced with the median count of 7. Since this variable represents a count, the imputed values were rounded and the column was converted to an integer data type.
- **Device type:** The 60 missing values were replaced with the category Unknown, preserving the records without assuming which device had been used.
- **Card type:** The 30 missing values were replaced with Unknown for the same reason: the original category could not be established from the available data.

The missing-value counts were recorded before imputation to document the extent of the issue. After these operations, the notebook reported **zero missing values across all columns**. The dataset therefore retained all 50,000 records rather than removing rows because of incomplete information.

7.3 Duplicate Record Checks

Duplicate records were examined to determine whether repeated observations could distort the analysis. Two checks were performed: one for exact duplicate rows and another for repeated values in Transaction_ID. Missing transaction identifiers were also checked.

The results showed **zero exact duplicate rows, zero duplicate transaction IDs, and zero missing transaction IDs**. Consequently, no records needed to be removed because of duplication. This helped preserve the original number of observations while confirming that the dataset contained no duplicates under the checks performed.

7.4 Validation of Binary Variables

The binary variables were examined to verify that their values followed the expected 0/1 format. The columns checked were IP_Address_Flag, Previous_Fraudulent_Activity, Is_Weekend, and Fraud_Label.

The observed values in these columns were consistent with the expected binary format. A further validation check identified **zero invalid binary values** across the four columns. For the fraud label, 0 represented a non-fraudulent transaction and 1 represented a fraudulent transaction. This check helped ensure that the variables could be used consistently when comparing fraudulent and non-fraudulent transactions during EDA.

7.5 Numerical Data Validation

Numerical variables were examined using descriptive statistics, including the mean, median, standard deviation, minimum, and maximum. The validation covered transaction amounts, account balances, daily transaction counts, average transaction amounts over seven days, failed transaction counts over seven days, card age, transaction distance, and risk scores.

Specific checks were performed to identify negative transaction amounts, negative transaction counts, and risk scores outside the expected range of 0 to 1. The results showed no negative transaction amounts, no negative values in the checked transaction-count columns, and no risk scores outside the 0–1 range.

The observed ranges included transaction amounts from 0 to 1,174.14, account balances from 500.48 to 99,998.31, card ages from 1 to 239, and transaction distances from 0.25 to 4,999.93. Daily transaction counts ranged from 1 to 14, while failed transaction counts over seven days ranged from 0 to 4. These checks did not identify values outside the validation rules applied in the notebook.

7.6 Investigation of Potential Outliers

Potential outliers were investigated using the interquartile range (IQR) method. The IQR is calculated as the difference between the third quartile (Q3) and the first quartile (Q1). Values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ are flagged as potential outliers for further examination.

For Transaction_Amount, the first quartile was 28.71 and the third quartile was 138.5825, producing an IQR of 109.8725. The resulting upper limit was approximately 303.39. A total of **2,268 transactions** were flagged as potential outliers, with transaction amounts ranging from 303.43 to 1,174.14 among the flagged records.

The same method was applied to Account_Balance, Avg_Transaction_Amount_7d, and Transaction_Distance. No potential outliers were identified in these three columns using the calculated IQR limits.

Potential outliers were investigated rather than automatically removed. A transaction amount that is unusually high is not necessarily incorrect or fraudulent; removing such records without further evidence could exclude meaningful observations from the fraud analysis.

7.7 Categorical Data Consistency Checks

The categorical variables Transaction_Type, Device_Type, Location, Merchant_Category, Card_Type, and Authentication_Method were examined to review their category values and frequencies. Additional checks were performed to identify leading or trailing spaces in the text values.

The notebook found **zero values with extra leading or trailing spaces** in the checked categorical columns. Missing device and card categories had already been represented as Unknown. These checks helped confirm that the category labels were consistently formatted for grouping and comparison during visualisation.

7.8 Timestamp and Transaction-Specific Checks

The Timestamp column was reviewed using sample records and then converted from text to a datetime data type using Pandas. The conversion used the format %m/%d/%Y %H:%M, with invalid values set to missing so they could be counted and checked.

The conversion produced **zero invalid or missing timestamps**. This confirmed that the timestamp values could be used for time-based analysis, including the examination of monthly transaction volumes and monthly fraud rates.

Two transactions with a transaction amount of zero were also identified. Both records were retained pending confirmation of whether zero-value transactions were permitted under the dataset's definition. In addition, checks confirmed that the daily and failed transaction-count columns contained no non-whole-number values.

7.9 Final Validation and Export of the Cleaned Dataset

After the cleaning and validation steps, the dataset was checked again for its dimensions, missing values, duplicate rows, and fraud-label values. The final checks reported **50,000 rows, 21 columns, zero missing values, zero duplicate rows, and fraud-label values of 0 and 1**.

The cleaned DataFrame was then exported to a CSV file named Capstone_Cleaned_Dataset.csv. The file was downloaded from Google Colab for use in the subsequent exploratory data analysis and visualisation stages.

Overall, the preprocessing stage addressed missing information, verified key data formats and ranges, investigated potential outliers, and confirmed the dataset's readiness for further analysis. The original number of records was preserved, and potential outliers and zero-value transactions were retained where there was insufficient evidence to justify their removal.

8. Exploratory Data Analysis (EDA)

8.1 Introduction to Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the distribution of transactions, identify patterns associated with fraudulent activity, compare fraud rates across different transaction characteristics, and examine relationships between numerical variables. The analysis was conducted using the cleaned dataset containing 50,000 transaction records.

Python libraries, including Pandas, Matplotlib, Seaborn, and Plotly, were used to calculate summary statistics and create visualisations. The analysis focused on the distribution of fraud labels, transaction amounts, risk scores, failed transaction counts, account balances, categorical transaction characteristics, monthly trends, and correlations between numerical variables.

Fraud rates were calculated as the number of fraudulent transactions divided by the total number of transactions in the relevant group, multiplied by 100. This measure was used alongside transaction counts to make comparisons between groups more meaningful.

8.2 Distribution of Fraudulent and Non-Fraudulent Transactions

The overall distribution of the target variable, Fraud_Label, was examined to understand the proportion of fraudulent and non-fraudulent transactions in the dataset.

Table 8.1: Distribution of transactions by fraud label

Fraud Classification	**Number of Transactions**	**Percentage**
Non-fraudulent (0)	33,933	67.87%
Fraudulent (1)	16,067	32.13%
Total	50,000	100.00%

However, fraudulent transactions also represented a substantial proportion of the dataset. This provided a useful basis for investigating how transaction characteristics differed between the two groups. The distribution describes the labels within this dataset and should not automatically be interpreted as the prevalence of fraud in real-world banking transactions.

8.3 Analysis of Numerical Transaction Characteristics

8.3.1 Distribution of Transaction Amounts

The distribution of Transaction_Amount was analysed to understand typical transaction values and the presence of unusually large transactions.

Table 8.2: Summary statistics for transaction amounts

Statistical Measure	**Value**
Mean	99.31
Median	69.61
Minimum	0.0
Maximum	1,174.14
Transactions below 100	31,573(63.15%)

The mean transaction amount of 99.31 was higher than the median of 69.61. This difference, together with the histogram, indicated a right-skewed distribution, where most transactions were concentrated at lower amounts and a smaller number of transactions had substantially higher values.

A total of 31,573 transactions, equivalent to 63.15% of the dataset, had amounts below 100. Although some transactions had relatively high values, a high transaction amount alone does not establish that a transaction was fraudulent or erroneous. These observations require consideration alongside other transaction characteristics.

8.3.2 Risk Score Distribution by Fraud Label

The distribution of Risk_Score was compared between fraudulent and non-fraudulent transactions. The visualisation showed that fraudulent transactions were generally concentrated at higher risk scores than non-fraudulent transactions.

This pattern suggests that the risk score may be useful for distinguishing between the two groups within the dataset. However, the visual comparison alone does not establish the accuracy of the

risk score or demonstrate how well it would identify previously unseen fraudulent transactions. Further validation would be required to evaluate its performance.

8.3.3 Failed Transaction Count by Fraud Status

The number of failed transactions recorded over the previous seven days, represented by Failed_Transaction_Count_7d, was compared across the two fraud-label groups.

The visualisation showed higher failed transaction counts among fraudulent transactions compared with non-fraudulent transactions. This observation was consistent with the positive correlation between failed transaction count and the fraud label discussed in Section 8.7.

Repeated failed transactions may warrant further investigation in a banking fraud-monitoring context. Nevertheless, the observed association does not establish that failed transactions cause fraud, and the count should not be used as a standalone indicator without further evaluation.

8.3.4 Account Balance Distribution by Fraud Status

Account balances were compared between fraudulent and non-fraudulent transactions to determine whether the two groups showed noticeable differences.

The visualisation indicated that the account balance distributions had broadly similar central values. This suggests that account balance alone did not show a strong visual distinction between the two groups in this dataset.

Consequently, account balance should be interpreted alongside other variables rather than assumed to be a strong standalone indicator of fraudulent activity.

8.4 Comparison of Fraud Rates Across Categorical Variables

Fraud rates were compared across locations, transaction types, card age groups, card types, authentication methods, and previous fraudulent activity. These comparisons helped identify whether certain categories showed noticeably different proportions of fraudulent transactions.

8.4.1 Fraud Rate by Location

The fraud rate was calculated for each of the five locations represented in the dataset.

Table 8.3: Fraud rate by location

Location	Total Transactions	Fraud Rate
London	9,945	31.99%
Mumbai	9,994	31.58%
New York	9,915	32.49%
Sydney	9,938	32.14%
Tokyo	10,208	32.47%

The fraud rates across the five locations were relatively similar, ranging from 31.58% in Mumbai to 32.49% in New York. Tokyo also recorded a relatively high rate of 32.47%.

The small differences suggest that location did not create a substantial separation in fraud rates in this dataset. These results should not be interpreted as evidence that the locations have identical real-world fraud risks, since the dataset's composition and generation process may affect the observed rates.

8.4.2 Fraud Rate by Transaction Type

The transaction types were compared to determine whether the proportion of fraudulent transactions varied according to the type of transaction.

Table 8.4: Fraud rate by transaction type

Transaction Type	Total Transactions	Fraudulent Transactions	Fraud Rate
ATM Withdrawal	12,453	4,046	32.49%
Bank Transfer	12,452	3,995	32.08%
Online	12,546	4,031	32.13%
POS	12,549	3,995	31.84%

ATM withdrawals recorded the highest fraud rate at 32.49%, while POS transactions recorded the lowest at 31.84%. The difference between the highest and lowest rates was 0.65 percentage points.

Overall, the fraud rates were closely grouped across the four transaction types. The results do not indicate a large difference in fraud rates by transaction type within this dataset.

8.4.3 Fraud Rate by Card Age Group

Card age was grouped into four intervals to compare the fraud rates for transactions associated with cards of different ages.

Table 8.5: Fraud rate by card age group

Card Age Group	Total Transactions	Fraudulent Transactions	Fraud Rate
1-60 days	12,566	4,040	32.15%
61-120 days	12,542	4,067	32.43%
121-180 days	12,569	4,037	32.12%
181-239 days	12,323	3,923	31.83%

The 61–120-day group had the highest fraud rate at 32.43%, while the 181–239-day group had the lowest at 31.83%. The difference between the two rates was 0.60 percentage points.

The rates remained relatively similar across all four groups. Therefore, the analysis did not reveal a substantial difference in fraud rates based solely on the card age groups used.

8.4.4 Fraud Rate by Card Type

Fraud rates were also compared across the card types recorded in the dataset.

Table 8.6: Fraud rate by card type

Card Type	Total Transactions	Fraudulent Transactions	Fraud Rate
Amex	12,415	4,019	32.37%
Discover	12,320	4,000	32.47%
Mastercard	12,681	4,036	31.83%
Visa	12,554	3,999	31.85%
Unknown	30	13	43.33%

Among the four identified card types, Discover recorded the highest fraud rate at 32.47%, followed by Amex at 32.37%. Mastercard and Visa recorded rates of 31.83% and 31.85%, respectively.

The Unknown category had a higher observed rate of 43.33%, but it contained only 30 transactions, of which 13 were fraudulent. This small sample means the rate is less stable and should not be compared directly with the much larger card-type groups without caution.

The results showed relatively similar fraud rates among the four identified card types. The higher rate for the Unknown category should be treated as an observation requiring further investigation rather than a conclusive finding.

8.4.5 Fraud Rate by Authentication Method

The relationship between authentication method and fraud rate was examined across biometric, OTP, PIN, and password-based authentication.

Table 8.7: Fraud rate by authentication method

Authentication Method	Total Transactions	Fraudulent Transactions	Fraud Rate
Biometric	12,591	4,038	32.07%
OTP	12,366	4,021	32.52%
PIN	12,586	3,978	31.61%
Password	12,457	4,030	32.35%

OTP authentication recorded the highest fraud rate at 32.52%, whereas PIN authentication recorded the lowest at 31.61%. The difference was 0.91 percentage points.

Although small differences were observed, all four authentication methods had fraud rates within a relatively narrow range. The results therefore do not demonstrate a substantial difference in fraud rates by authentication method in this dataset. They also do not establish that any authentication method is inherently more or less secure.

8.4.6 Fraud Rate by Previous Fraudulent Activity

The Previous_Fraudulent_Activity variable was compared with the current fraud label to examine whether transactions associated with previous fraudulent activity showed a higher current fraud rate.

Table 8.8: Fraud rate by previous fraudulent activity

Previous Fraudulent Activity	Total Transactions	Fraud Rate
No previous fraud	45,080	32.15%
Previous fraud	4,920	32.03%

The fraud rate was 32.15% for transactions associated with no previous fraudulent activity and 32.03% for transactions associated with previous fraudulent activity. The difference was only 0.12 percentage points, with the group without previous fraudulent activity showing the slightly higher rate.

This result does not show a higher current fraud rate for the group marked as having previous fraudulent activity. The variable may still be relevant when combined with other information, but its relationship with the current fraud label was limited in this comparison.

8.5 Fraud Rate by Risk Category

To compare fraud rates across levels of transaction risk, the continuous Risk_Score variable was divided into three categories: Low, Medium, and High. The categories were defined using the following thresholds:

- **Low:** Risk score below 0.33
- **Medium:** Risk score from 0.33 to below 0.66
- **High:** Risk score of 0.66 or above

The fraud rate was then calculated for each category.

Table 8.9: Fraud rate by risk category

Risk Category	**Total Transactions**	**Fraudulent Transactions**	**Fraud Rate**
Low	16,343	3,266	19.98%
Medium	16,647	3,272	19.66%
High	17,010	9,529	56.02%

The High risk category recorded a fraud rate of 56.02%, substantially higher than the Low category at 19.98% and the Medium category at 19.66%. The difference between the High and Low categories was 36.04 percentage points.

The Medium category had a slightly lower fraud rate than the Low category. Therefore, the observed results show a marked difference for the High category rather than a consistent increase in fraud rates across all three categories.

This pattern indicates that the existing risk score is associated with the fraud label in the dataset. However, because the score was already present in the source dataset, its original calculation and validation process are not established by this analysis. The observed association should not be treated as independent proof of predictive accuracy.

8.6 Temporal Analysis of Transactions and Fraud Rates

8.6.1 Monthly Transaction Volume

The number of transactions recorded in each month of 2023 was analysed to understand how transaction volume varied over time.

Table 8.10: Monthly transaction volume

Month	Transactions
January	4,188
February	3,903
March	4,259
April	4,106
May	4,162
June	4,160
July	4,213
August	4,384
September	4,087
October	4,164
November	4,085
December	4,289

August recorded the highest transaction volume, with 4,384 transactions, while February recorded the lowest, with 3,903 transactions. Transaction volumes fluctuated across the year but remained within a relatively narrow range.

The observed variation indicates changes in monthly transaction activity within the dataset. However, the chart alone does not explain the reasons for these fluctuations or establish a recurring seasonal pattern

8.6.2 Monthly Fraud Rate Trend

Monthly fraud rates were calculated separately from monthly transaction volumes to determine whether the proportion of fraudulent transactions changed throughout the year.

Table 8.11: Monthly fraud rate

Month	Fraud rate
January	33.09%
February	31.92%
March	31.93%
April	31.90%
May	31.60%
June	32.38%
July	32.14%
August	32.71%
September	31.20%
October	32.25%
November	31.85%
December	32.55%

January recorded the highest monthly fraud rate at 33.09%, while September recorded the lowest at 31.20%. The monthly rates varied by 1.89 percentage points between these two extremes.

The fraud rate fluctuated across the year without a consistent upward or downward trend. Monthly transaction volume and monthly fraud rate therefore provided different perspectives: the month with the highest transaction volume was August, whereas the highest fraud rate occurred in January.

8.7 Relationship Between Numerical Variables

8.7.1 Transaction Amount and Risk Score

A scatter plot was used to examine the relationship between Transaction_Amount and Risk_Score. This visualisation helped assess whether higher transaction amounts were consistently associated with higher risk scores or whether the observations were more widely distributed.

The scatter plot should be interpreted as a visual assessment of the relationship between the two variables. A transaction amount on its own does not establish the risk associated with a transaction, and the presence or absence of a clear visual pattern does not prove causation.

8.7.2 Correlation Analysis

A correlation heatmap was generated to examine the direction and strength of linear relationships between numerical variables, including their relationships with Fraud_Label.

Table 8.12: Key correlations with the fraud label

Variable	**Correlation with Fraud Label**
Failed transaction count for 7 days	+0.51
Risk Score	+0.39
Other numerical variables examined	Close to zero

The strongest positive correlation with the fraud label was observed for Failed_Transaction_Count_7d, at +0.51. This indicates a moderate positive linear association: higher failed transaction counts tended to be associated with the fraud label being 1.

Risk_Score also showed a positive correlation with the fraud label, at +0.39. This was consistent with the earlier risk-category analysis, in which the High risk category had a substantially higher fraud rate than the Low and Medium categories.

The other numerical variables examined had correlations close to zero with the fraud label, indicating limited linear association in this analysis. A near-zero correlation does not necessarily rule out nonlinear relationships or relationships that depend on combinations of variables.

Correlation does not establish causation. These findings describe associations within the dataset and do not independently establish that a variable causes fraud or can accurately predict future fraudulent transactions.

8.8 Summary of Key EDA Findings

The EDA identified several patterns that provide a foundation for further investigation.

Table 8.13: Summary of highlighted findings

Area analyzed	Key finding	Interpretation
Overall fraud distribution	16,067 of 50,000 transactions were labelled fraudulent (32.13%).	Fraudulent transactions formed a substantial proportion of this dataset.
Transaction amount	Mean: 99.31; median: 69.61; 63.15% of transactions were below 100.	Transaction amounts were right-skewed, with a smaller number of high-value observations.
Failed transaction count	Correlation with fraud label: +0.51.	Higher failed transaction counts were associated with the fraud label.
Account balance	Similar central values across fraud groups.	Account balance did not show a strong visual distinction on its own.
Risk category	High: 56.02%; Low: 19.98%; Medium: 19.66%.	The High category showed a substantially higher fraud rate.
Previous fraudulent activity	No previous fraud: 32.15%; previous fraud: 32.03%.	The group with previous fraudulent activity did not have a higher observed fraud rate.

Monthly activity	August had the highest transaction volume; January had the highest fraud rate.	Transaction volume and fraud rate followed different monthly patterns.
Correlation analysis	Risk score correlated +0.39 with the fraud label.	Risk score was positively associated with the fraud label, but its predictive performance was not independently tested.

Overall, the EDA highlighted failed transaction activity and risk score as variables associated with the fraud label. In contrast, account balance and the categorical characteristics examined showed more limited separation in this dataset. The monthly analysis revealed fluctuations in both transaction volume and fraud rate without a clear consistent trend.

These findings are descriptive and should be interpreted within the limits of the available dataset. They can help identify variables for further investigation, but they do not establish causation, guarantee predictive accuracy, or demonstrate that the same patterns would occur in other banking datasets.

9. Data Visualization

A total of **[insert final number] visualisations** were created to present the findings of the exploratory data analysis (EDA) clearly and effectively. The visualisations were developed using Python libraries, including **Matplotlib, Seaborn, and Plotly**, with each library supporting the presentation and interpretation of the dataset. Different chart types were used, including bar charts, histograms, box plots, count plots, scatter plots, line charts, and correlation heatmaps, where applicable.

The visualisations were organised into three categories based on the number of variables analysed:

- **Univariate Analysis:** Examined individual variables to understand their distributions, frequencies, and overall characteristics. Examples include histograms, count plots, and box plots.
- **Bivariate Analysis:** Examined relationships between two variables, such as comparing transaction amounts or risk scores across fraudulent and non-fraudulent transactions. Examples include grouped bar charts, scatter plots, and line charts.

- **Multivariate Analysis:** Examined relationships involving multiple variables to identify broader patterns and associations. Correlation heatmaps and other multi-variable comparisons were used where appropriate.

This classification helped structure the visual analysis and made it easier to interpret transaction patterns, compare fraud-related characteristics, and identify trends within the dataset. The following sections provide a brief explanation of each visualisation, including its purpose and key findings.

10.1 Distribution of Fraudulent and Non-Fraudulent Transactions

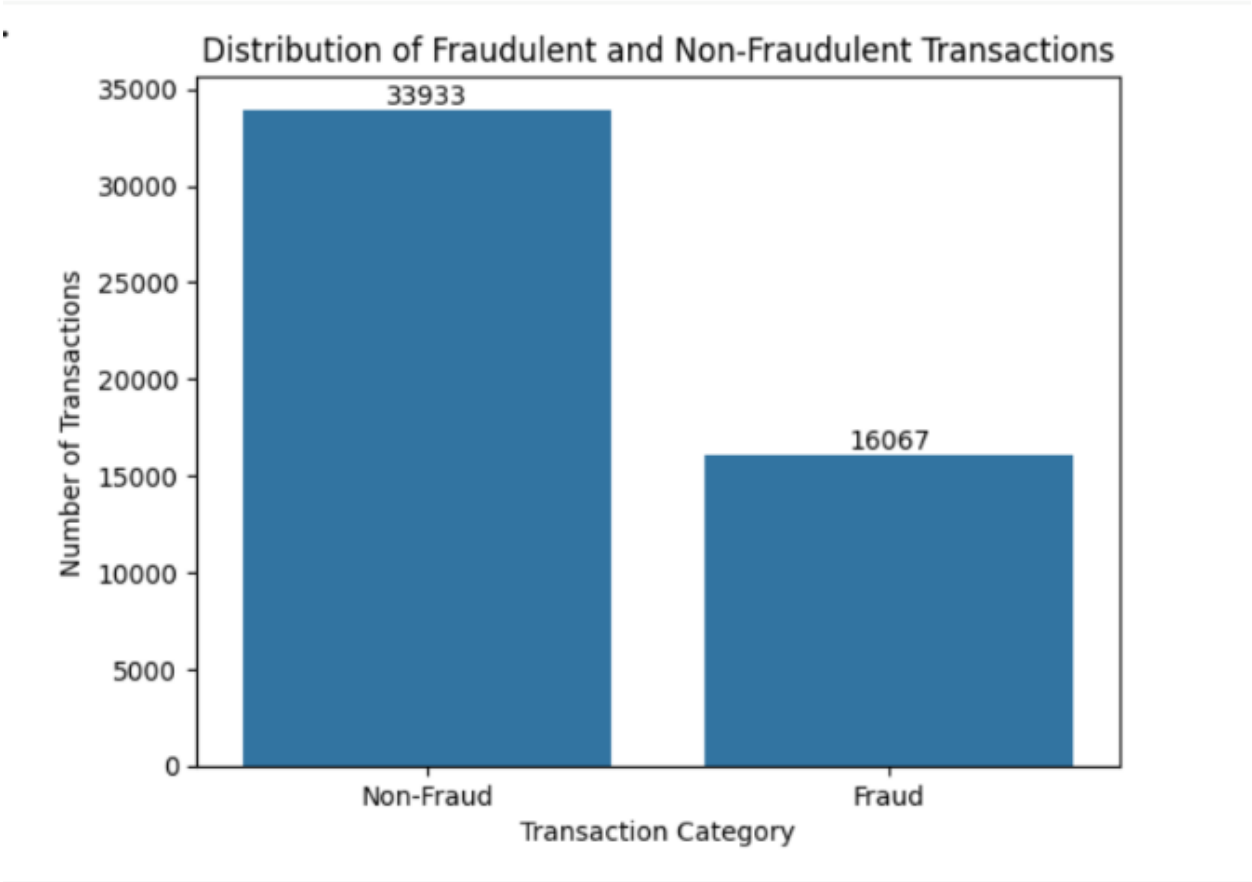


Fig 10.1 Distribution of Fraudulent and non Fraudulent transactions

Library Used: Seaborn

Chart type : Count Plot

Analysis Type: Univariate Analysis

Purpose: This bar chart shows the distribution of fraudulent and non-fraudulent transactions in the dataset.

Observation: Out of 50,000 transactions, 33,933 (67.87%) were non-fraudulent, while 16,067 (32.13%) were fraudulent.

Interpretation: Non-fraudulent transactions formed the majority of the dataset. However, fraudulent transactions accounted for approximately one-third of all transactions, highlighting the importance of further fraud analysis.

10.2 Distribution of Transaction Amounts

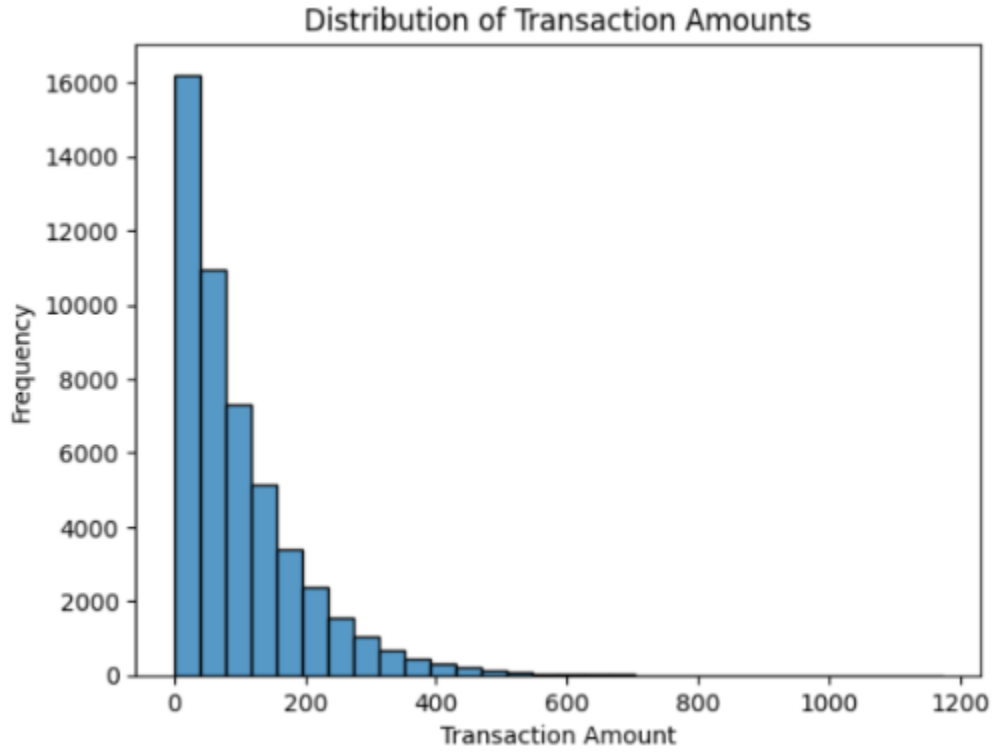


Figure 10.2 Distribution of Transaction Amounts

Chart Type: Histogram (30 bins)

Library Used: Seaborn (sns.histplot), with Matplotlib used for the title, axis labels, and display.

Analysis Type: Univariate Analysis

Purpose: This histogram shows the frequency distribution of transaction amounts and helps identify the range in which most transactions occur.

Observation: The distribution is right-skewed, with most transactions concentrated at lower amounts and fewer transactions occurring at higher amounts. The mean transaction amount was 99.31, while the median was 69.61. Transaction amounts ranged from 0 to 1,174.14.

Interpretation: The mean being higher than the median supports the right-skewed pattern, as higher-value transactions pull the mean upward. The histogram helps illustrate the typical transaction amount and the presence of relatively high-value transactions

10.3 **Risk Score Distribution by Fraud Label**

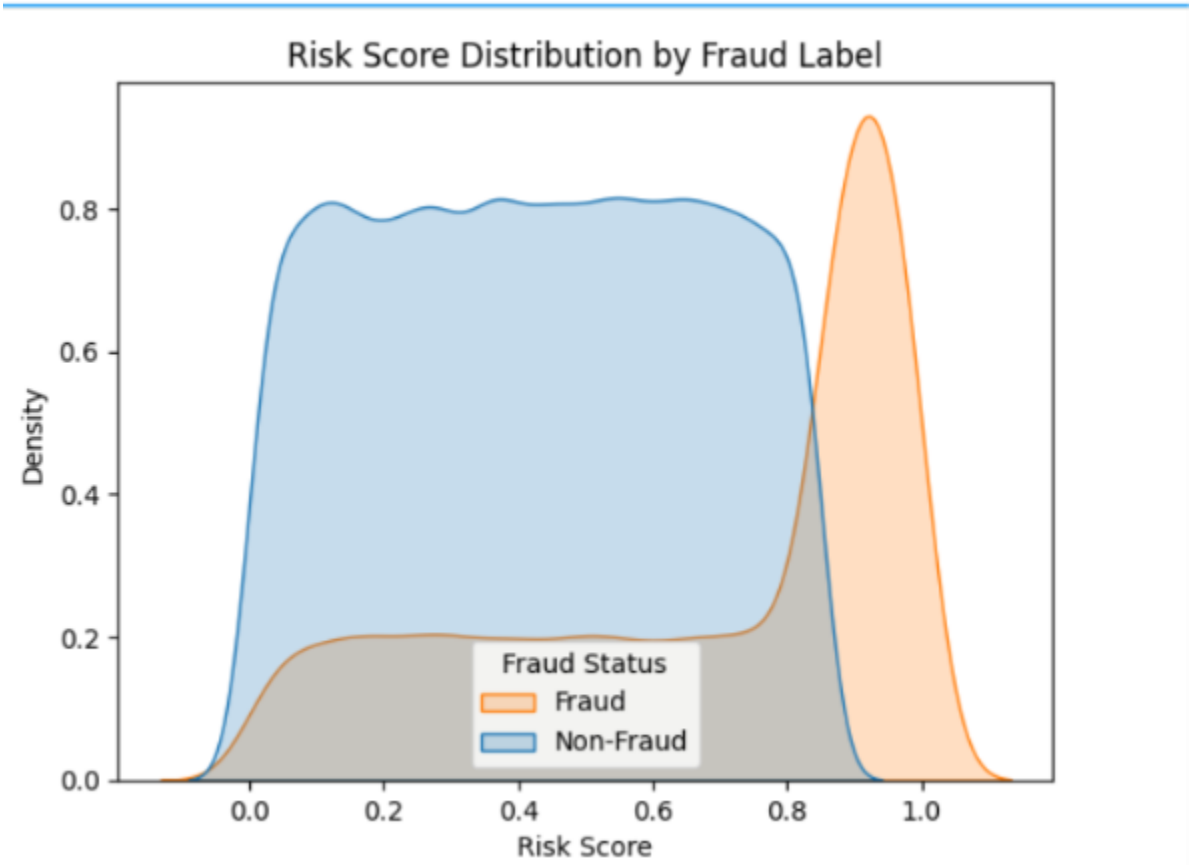


Fig 10.3 Risk Score Distribution by Fraud label

Chart Type: Kernel Density Estimate (KDE) Plot

Library Used: Seaborn (sns.kdeplot), with Matplotlib used for the title, axis labels, and legend.

Analysis Type: Bivariate Analysis

Purpose: This plot compares the distribution of risk scores for fraudulent and non-fraudulent transactions to examine how the scores differ between the two groups.

Observation: Fraudulent transactions are concentrated at higher risk scores, particularly around 0.9–1.0, whereas non-fraudulent transactions are more widely distributed across lower and moderate scores.

Interpretation: The visualisation shows an association between higher risk scores and fraudulent transactions in this dataset. However, this pattern alone does not establish the accuracy of the risk score as a fraud detection tool.

10.4 Failed Transaction Count by Fraud Status

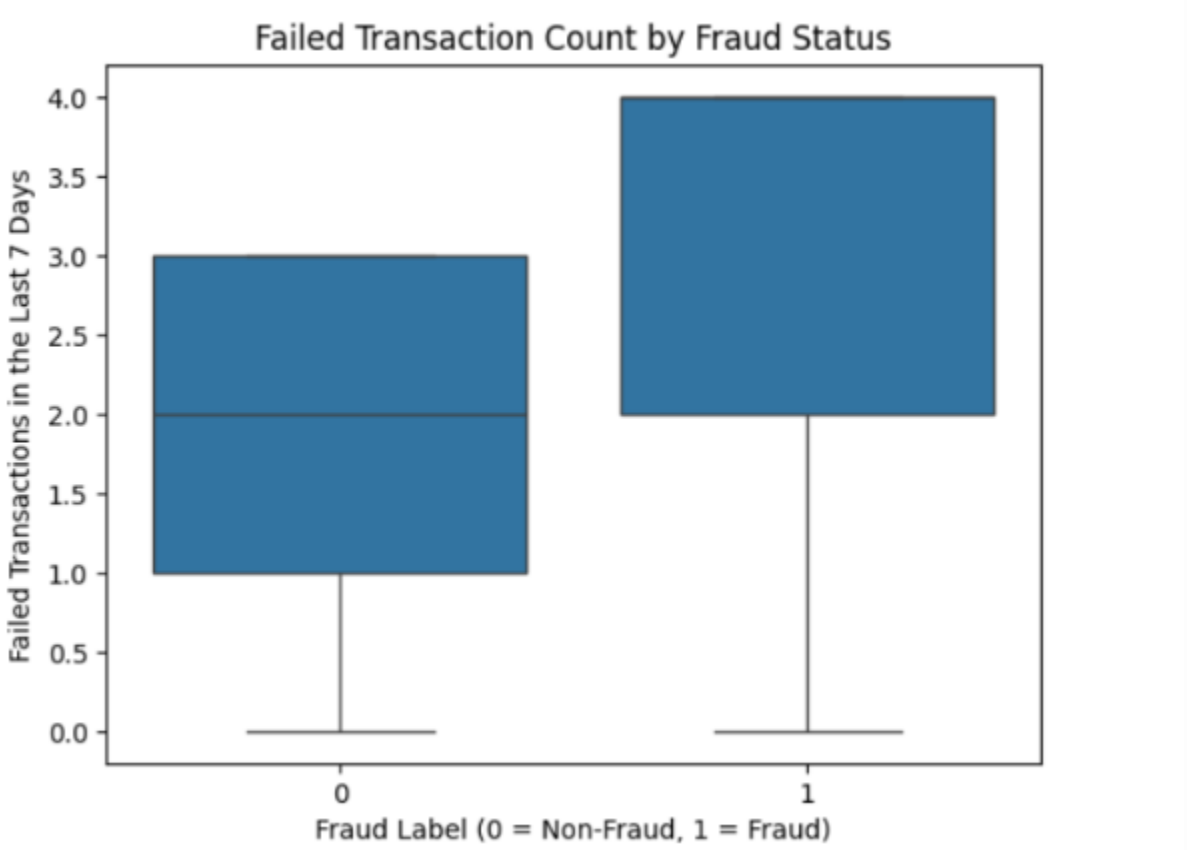


Fig 10.4 Failed Transaction Count by fraud status

Chart Type: Box Plot

Library Used: Seaborn (sns.boxplot), with Matplotlib used for the title and axis labels.

Analysis Type: Bivariate Analysis

Purpose: This box plot compares the distribution of failed transactions during the previous seven days between fraudulent and non-fraudulent transactions.

Observation: Non-fraudulent transactions have a median failed transaction count of 2, while the fraudulent group shows a higher median of approximately 3. The fraudulent group also has a higher upper quartile, indicating that its failed transaction counts tend to be higher.

Interpretation: The chart suggests an association between a higher number of recent failed transactions and fraudulent activity. This variable may be useful for further investigation, although the plot alone cannot establish causation or confirm predictive performance.

10.5 Account Balance by Fraud Status

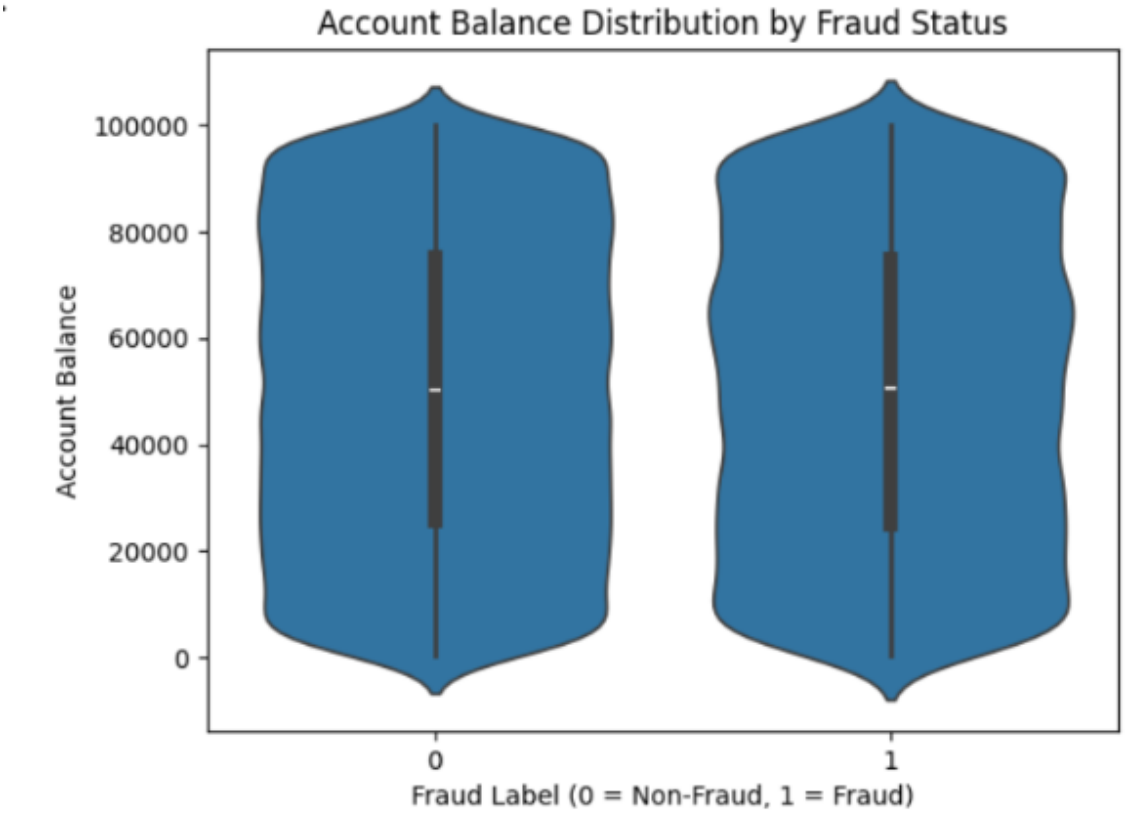


Fig 10.5 Account Balance by Fraud Status

Library Used: Seaborn (sns.violinplot), with Matplotlib used for the title and axis labels.

Analysis Type: Bivariate Analysis

Purpose: This plot compares the distribution of account balances between fraudulent and non-fraudulent transactions to examine whether account balance patterns differ by fraud status.

Observation: Both groups show broadly similar account balance distributions. The median account balance is approximately 50,000 in each group, and the overall ranges are also similar.

Interpretation: The visualisation does not show a clear difference in account balance between the two fraud categories. Therefore, account balance alone may not be sufficient to distinguish fraudulent from non-fraudulent transactions.

10.6 Fraud Rate by Location

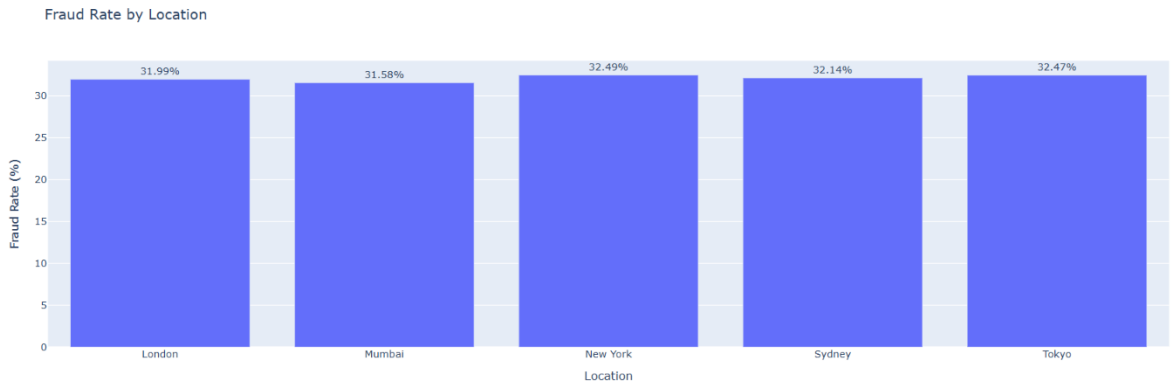


Fig 10.6 Fraud Rate by Location

Chart Type: Bar Chart

Library Used: Plotly (px.bar)

Analysis Type: Bivariate Analysis

Purpose: This chart compares the percentage of fraudulent transactions across five locations: London, Mumbai, New York, Sydney, and Tokyo.

Observation: New York recorded the highest fraud rate at 32.49%, closely followed by Tokyo at 32.47%. Mumbai had the lowest rate at 31.58%. London and Sydney recorded rates of 31.99% and 32.14%, respectively.

Interpretation: Fraud rates were relatively similar across all five locations, ranging from 31.58% to 32.49%. The small differences suggest that location alone may not strongly distinguish fraudulent transactions in this dataset.

10.7 Fraud by Transaction Type

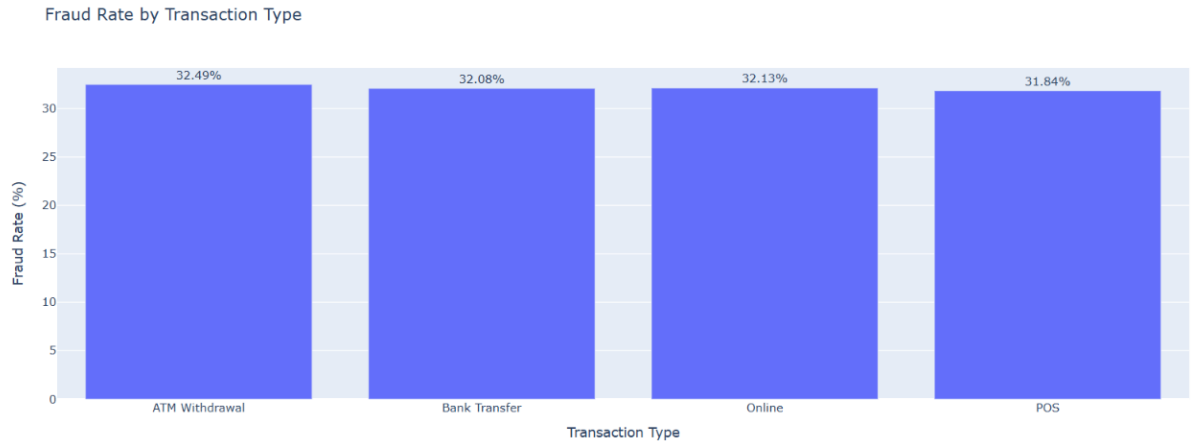


Fig 10.6 Fraud by Transaction Type

Chart Type: Bar Chart

Library Used: Plotly (px.bar)

Analysis Type: Bivariate Analysis

Purpose: This chart compares the percentage of fraudulent transactions across four transaction types: ATM Withdrawal, Bank Transfer, Online, and POS (Point of Sale).

Observation: ATM Withdrawal recorded the highest fraud rate at 32.49%, followed by Online transactions at 32.13%, Bank Transfer at 32.08%, and POS transactions at 31.84%, which had the lowest rate.

Interpretation: Fraud rates were relatively similar across all four transaction types, with a difference of less than one percentage point between the highest and lowest rates. Transaction type alone therefore shows limited variation in fraud rates within this dataset.

10.8 Fraud Rate by Card Age Group

Chart Type: Bar Chart

Library Used: Plotly (px.bar)

Analysis Type: Bivariate Analysis

Purpose: This chart compares fraud rates across four card age groups: 1–60 days, 61–120 days, 121–180 days, and 181–239 days.

Observation: The 61–120 days group recorded the highest fraud rate at 32.43%, followed by the 1–60 days group at 32.15% and the 121–180 days group at 32.12%. The 181–239 days group had the lowest rate at 31.83%.

Interpretation: Fraud rates were relatively consistent across all four card age groups, with a difference of only 0.60 percentage points between the highest and lowest rates. This suggests that card age alone shows limited variation in fraud rates within this dataset.

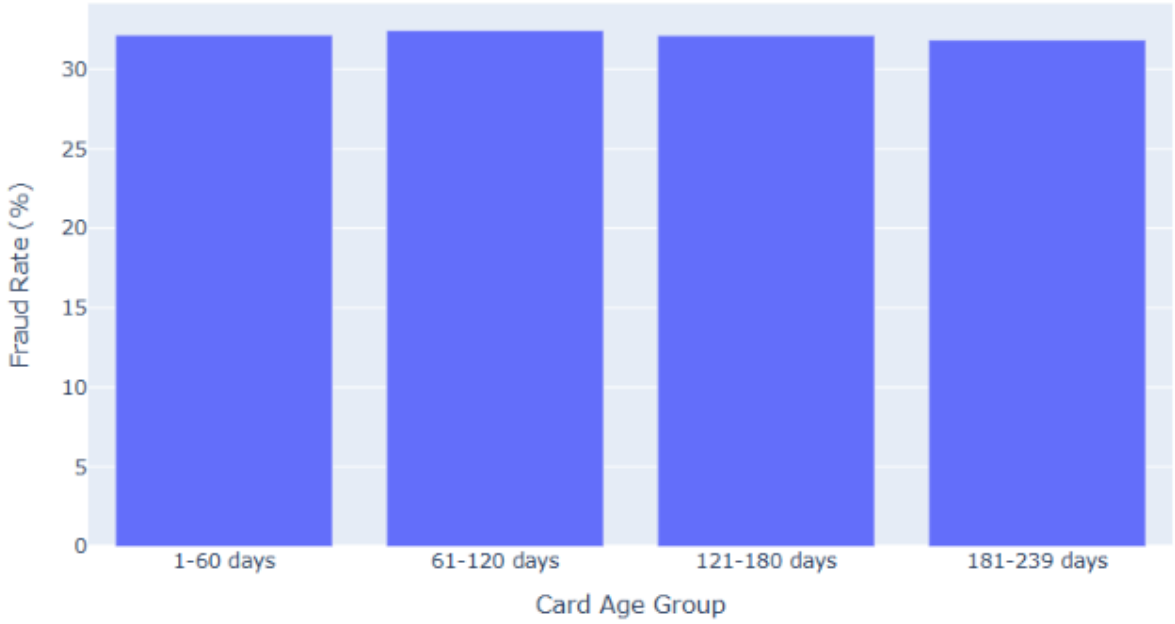


Fig 10.8 Fraud rate by card age group

10.9 Fraud Rate by Risk Category

Library Used: Plotly (px.bar)

Analysis Type: Bivariate Analysis

Purpose: This chart compares fraud rates across four card age groups: 1–60 days, 61–120 days, 121–180 days, and 181–239 days.

Observation: The 61–120 days group recorded the highest fraud rate at 32.43%, followed by the 1–60 days group at 32.15% and the 121–180 days group at 32.12%. The 181–239 days group had the lowest rate at 31.83%.

Interpretation: Fraud rates were relatively consistent across all four card age groups, with a difference of only 0.60 percentage points between the highest and lowest rates. This suggests that card age alone shows limited variation in fraud rates within this dataset.

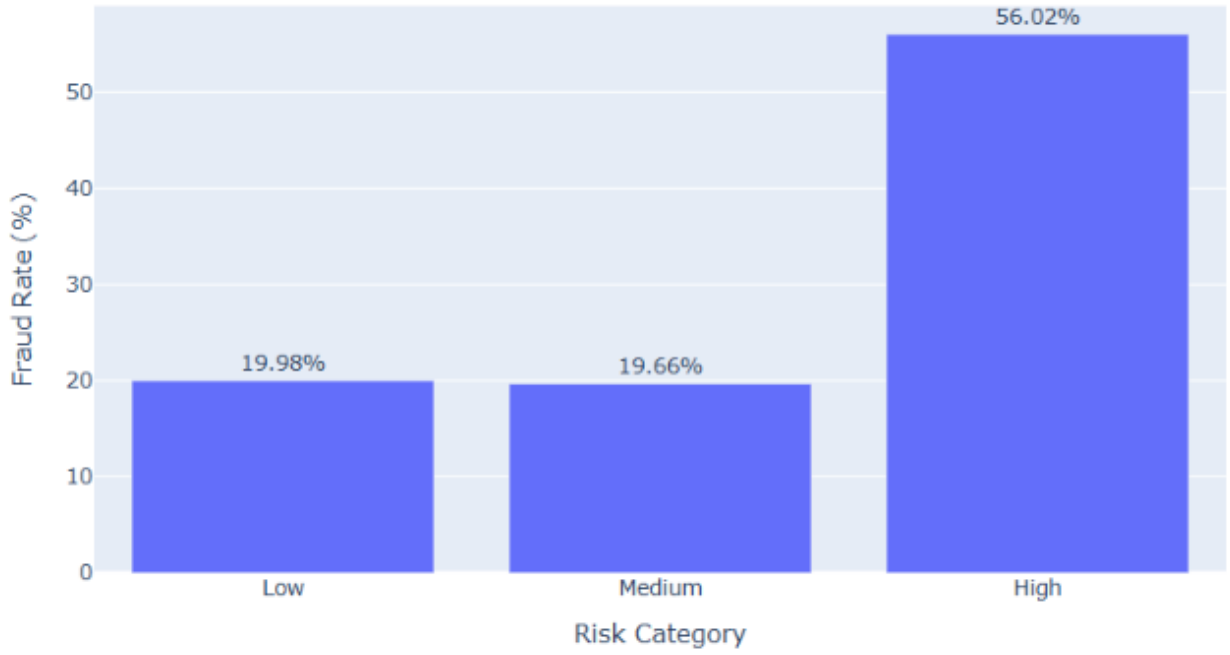


Fig 10.9 Fraud Rate by Risk Category

10.10 Monthly Transaction Trend



Fig 10.10 Monthly Transaction Trend

Chart Type: Line Chart with Markers

Library Used: Plotly (px.line)

Analysis Type: Univariate Time-Based Analysis

Purpose: This line chart shows the monthly number of transactions recorded during 2023, helping identify variations in transaction volume throughout the year.

Observation: August recorded the highest transaction volume, with 4,384 transactions, while February had the lowest, with 3,903 transactions. Monthly transaction counts fluctuated throughout the year, and December recorded 4,289 transactions.

Interpretation: Transaction volumes varied from month to month, with a noticeable increase in August and a decline in February. However, the chart does not show a consistent upward or downward trend across the year.

10.11 Transaction Amount Vs Risk Score



Fig 10.11 Transaction Amount Vs Risk Score

Chart Type: Scatter Plot

Library Used: Seaborn (sns.scatterplot), with Matplotlib used for the title and axis labels.

Analysis Type: Multivariate Analysis

Purpose: This scatter plot examines the relationship between transaction amount and risk score, with colours representing fraud status (0 = non-fraud, 1 = fraud).

Observation: Fraudulent transactions (orange) are concentrated more heavily at higher risk scores, particularly around 0.9–1.0. Non-fraudulent transactions (blue) are distributed across a broader range of risk scores. Both categories appear across different transaction amounts, with many transactions concentrated below 500.

Interpretation: The plot suggests that fraud status is more visibly differentiated by risk score than by transaction amount alone. However, the overlapping points indicate that transaction amount by itself does not clearly separate fraudulent from non-fraudulent transactions. This visualisation shows associations rather than causation.

10.12 Correlation Heatmap

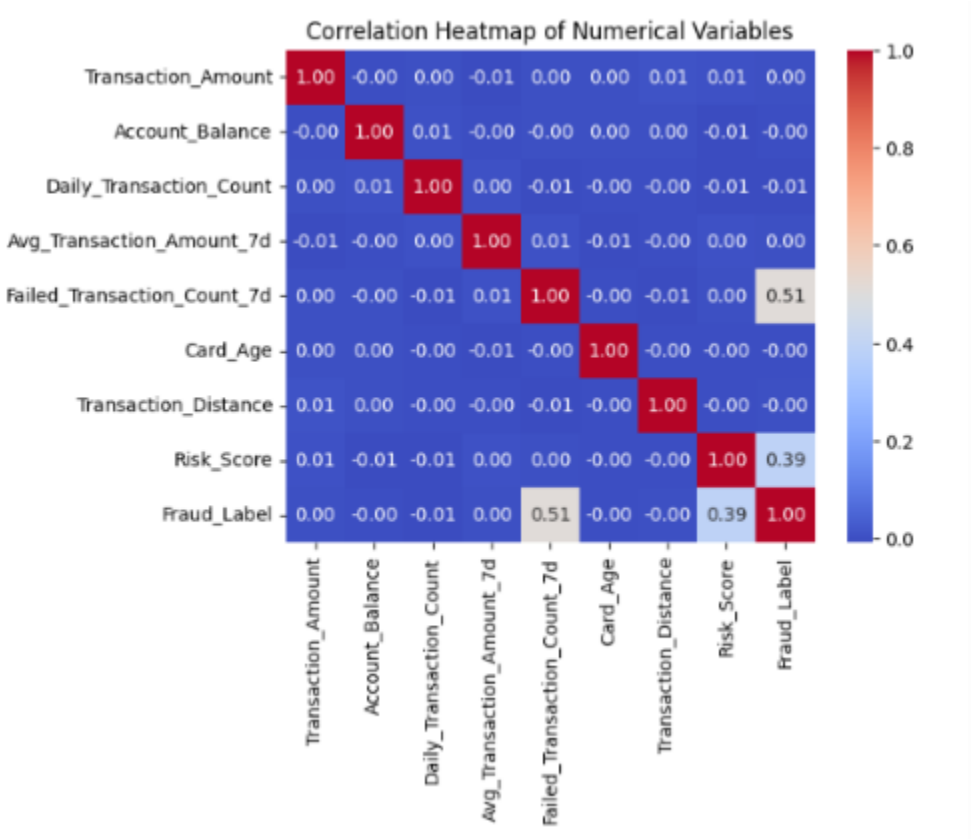


Fig 10.12 Correlation Heatmap

Chart Type: Correlation Heatmap

Library Used: Seaborn (sns.heatmap), with Matplotlib used for the title.

Analysis Type: Multivariate Analysis

Purpose: This heatmap displays the correlation coefficients between numerical variables in the dataset, including transaction amount, account balance, failed transaction count, risk score, and fraud label. It helps identify the strength and direction of linear relationships between variables.

Observation: The strongest positive correlations with Fraud_Label are observed for Failed_Transaction_Count_7d (0.51) and Risk_Score (0.39). Most other numerical variables have correlations close to zero with the fraud label. The diagonal values are 1.00 because each variable is perfectly correlated with itself.

Interpretation: The heatmap indicates that failed transaction count and risk score have the strongest positive linear associations with fraud status among the variables shown. However, correlation does not establish causation, and these results alone do not confirm how accurately the variables can predict fraud.

10.13 Fraud by Authentication Method

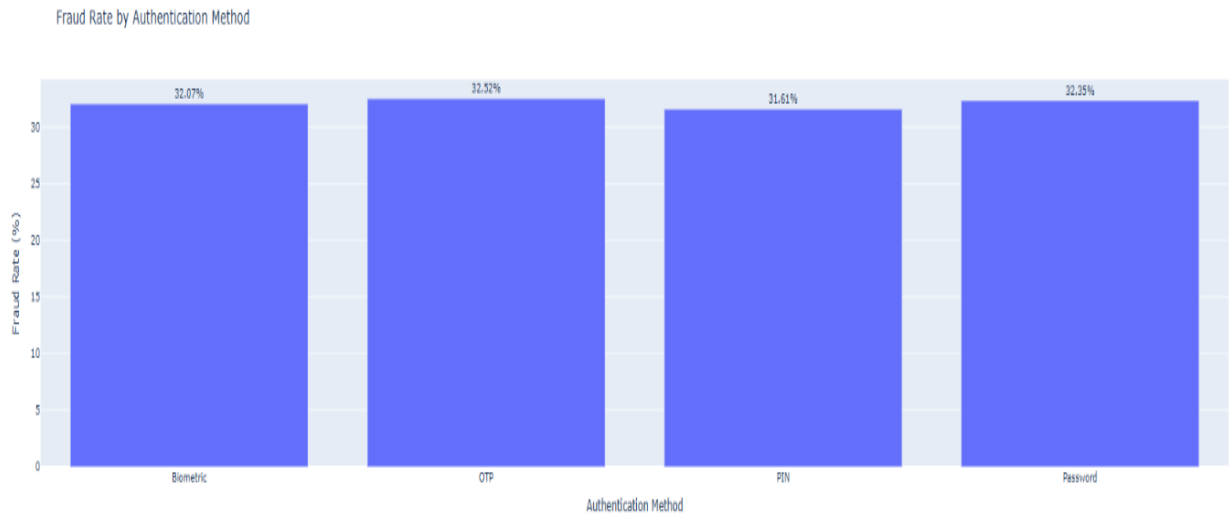


Fig 10.13 Fraud by Authentication Method

Chart Type: Bar Chart

Library Used: Plotly (px.bar)

Analysis Type: Bivariate Analysis

Purpose: This chart compares the percentage of fraudulent transactions across four authentication methods: Biometric, OTP, PIN, and Password.

Observation: OTP recorded the highest fraud rate at 32.52%, followed by Password at 32.35%, Biometric at 32.07%, and PIN at 31.61%, which had the lowest rate.

Interpretation: Fraud rates were relatively similar across all four authentication methods, with a difference of 0.91 percentage points between the highest and lowest rates. The results show limited variation in fraud rates by authentication method within this dataset, suggesting that this variable alone may not clearly distinguish fraudulent transactions.

10.14 Fraud Rate by Previous Fraudulent Activity



Fig 10.14 Fraud rate by previous fraudulent activity

Chart Type: Bar Chart

Library Used: Plotly (px.bar)

Analysis Type: Bivariate Analysis

Purpose: This chart compares fraud rates between transactions associated with accounts with and without previous fraudulent activity.

Observation: The fraud rate was 32.15% for transactions with no previous fraudulent activity and 32.03% for those with previous fraudulent activity. The dataset contained 45,080 transactions in the first group and 4,920 in the second group.

Interpretation: The fraud rates were almost identical, with only a 0.12 percentage-point difference. In this dataset, previous fraudulent activity does not show a substantial difference in the observed fraud rates between the two groups.

10.15 Monthly Fraud Rate Trend

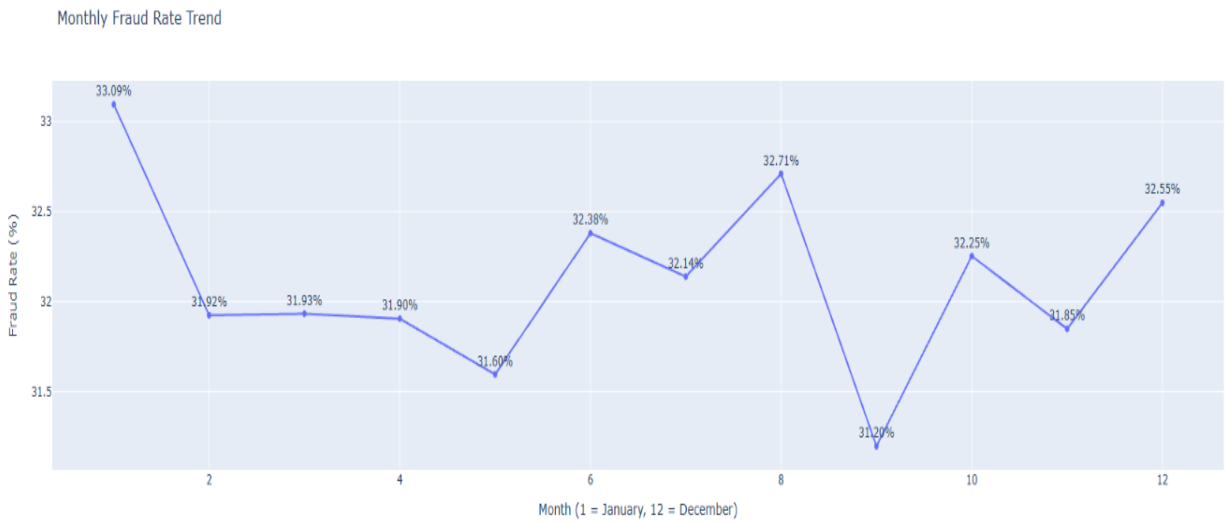


Fig 10.15 Monthly Fraud Rate Trend

Chart Type: Line Chart with Markers

Library Used: Plotly (px.line)

Analysis Type: Bivariate Analysis (Time and Fraud Rate)

Purpose: This chart illustrates how the percentage of fraudulent transactions changed across the 12 months of 2023.

Observation: January recorded the highest fraud rate at 33.09%, while September had the lowest at 31.20%. The rate increased to 32.71% in August, declined sharply in September, and rose again to 32.55% in December.

Interpretation: The fraud rate fluctuated throughout the year but remained within a relatively narrow range of 31.20% to 33.09%. No consistent upward or downward trend was observed, indicating that the monthly fraud rate was relatively stable overall despite short-term variations.

10.16 Fraud Rate by Card Type

Chart Type: Bar Chart

Library Used: Plotly (px.bar)

Analysis Type: Bivariate Analysis

Purpose: This chart compares fraud rates across different card types: Amex, Discover, Mastercard, Visa, and Unknown.

Observation: The Unknown category recorded the highest fraud rate at 43.33%, while Discover recorded 32.47%, Amex 32.37%, Visa 31.85%, and Mastercard 31.83%. However, the Unknown category contained only 30 transactions, compared with more than 12,000 transactions for each recognised card type.

Interpretation: Fraud rates were relatively similar across the four recognised card types. Although the Unknown category had a higher observed rate, its very small sample size means this result should be interpreted cautiously and should not be treated as a reliable indication of greater fraud risk.

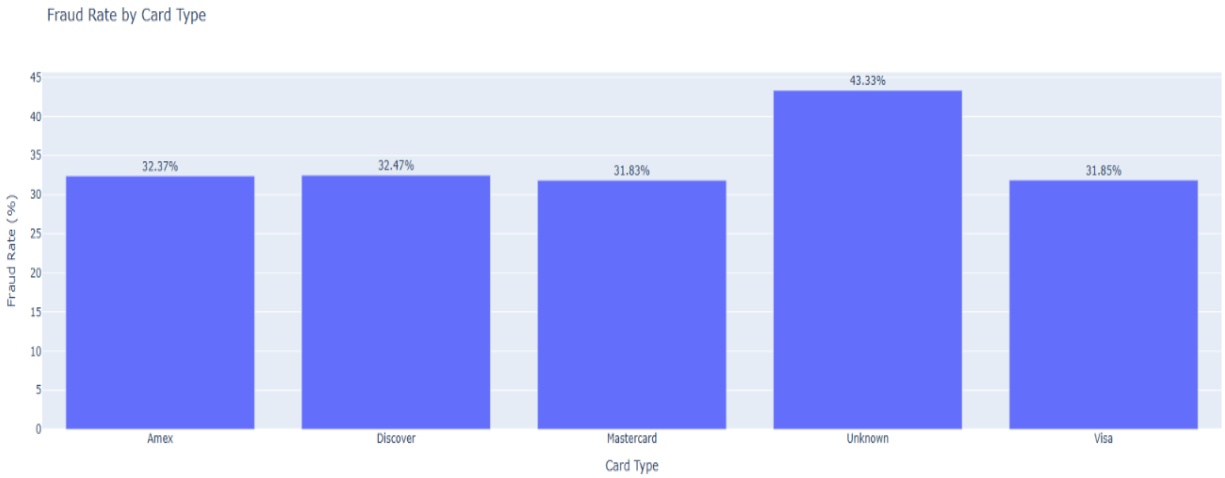


Fig 10.16 Fraud Rate by Card Type

11. Key Findings and Insights

The exploratory data analysis and visualisations revealed several patterns in the 50,000-transaction dataset. The findings highlight the distribution of fraudulent transactions, the relationship between risk scores and fraud, and the extent to which transaction characteristics differ between fraudulent and non-fraudulent transactions.

1. **Distribution of Fraudulent Transactions**

Of the 50,000 transactions analysed, 16,067 (32.13%) were labelled as fraudulent, while 33,933 (67.87%) were non-fraudulent. This indicates that fraudulent transactions represented a substantial proportion of the dataset and warranted further investigation.

2. **Risk Score and Fraud Status**

Risk score showed a clear association with fraud status. Transactions in the High Risk category had a fraud rate of 56.02%, compared with 19.98% in the Low Risk category and 19.66% in the

Medium Risk category. The correlation between Risk_Score and Fraud_Label was 0.39, indicating a positive relationship. These findings suggest that higher risk scores were associated with a greater proportion of fraudulent transactions in this dataset.

3. Failed Transaction Activity

The correlation heatmap showed a positive correlation of 0.51 between Failed_Transaction_Count_7d and Fraud_Label, making it the strongest positive correlation with the fraud label among the numerical variables examined. The box plot also showed a higher median failed transaction count for fraudulent transactions than for non-fraudulent transactions. This suggests that recent failed transaction activity may be a useful characteristic to investigate when analysing potential fraud.

4. Transaction Amount and Account Balance

Transaction amounts were right-skewed, with a mean of 99.31 and a median of 69.61. Most transactions involved lower amounts, while a smaller number had substantially higher values. The scatter plot showed that fraudulent and non-fraudulent transactions occurred across different transaction amounts, meaning transaction amount alone did not clearly separate the two groups. Account balance distributions were also broadly similar across fraud categories.

5. Differences Across Transaction Categories

Fraud rates were relatively similar across locations, transaction types, card age groups, authentication methods, and recognised card types. For example, fraud rates across locations ranged from 31.58% in Mumbai to 32.49% in New York. Similarly, rates across authentication methods ranged from 31.61% for PIN to 32.52% for OTP. These small differences suggest that the individual categories examined did not show substantial variation in fraud rates within this dataset.

The Unknown card type category had a higher observed fraud rate of 43.33%; however, it contained only 30 transactions. Therefore, this result should be interpreted cautiously because of its small sample size. Previous fraudulent activity also showed almost identical fraud rates: 32.15% for transactions without previous fraudulent activity and 32.03% for those with previous fraudulent activity.

6. Monthly Transaction and Fraud Trends

Transaction volumes fluctuated throughout 2023. August recorded the highest number of transactions, at 4,384, while February recorded the lowest, at 3,903. Monthly fraud rates ranged from 31.20% in September to 33.09% in January. Despite these fluctuations, no consistent upward or downward trend was observed in the monthly fraud rate.

Overall Insights

Overall, the analysis identified risk score and recent failed transaction activity as the numerical variables most strongly associated with fraud status in the dataset. In contrast, transaction amount, account balance, location, transaction type, authentication method, and card age showed limited ability to distinguish fraud categories through the descriptive comparisons performed.

These findings provide a basis for further fraud detection analysis. However, the observed associations do not establish causation or demonstrate predictive performance. Additional testing would be required to determine whether these characteristics can reliably identify fraudulent transactions in unseen data.

12. Recommendations

Based on the findings from the exploratory data analysis, the following recommendations are proposed to support more effective fraud monitoring and further development of fraud detection systems.

1. **Prioritise High-Risk Transactions**

Transactions assigned high risk scores should receive additional scrutiny, as the High Risk category recorded a fraud rate of 56.02%, compared with approximately 20% for the Low and Medium categories. Risk scores could be used to prioritise transactions for further review, subject to validation of their reliability and suitability for operational use.

2. **Monitor Failed Transaction Activity**

The positive correlation of 0.51 between failed transaction count and fraud status suggests that recent failed transaction activity deserves further investigation. Fraud monitoring systems could consider repeated failed transactions as one of several indicators for identifying transactions that may require additional verification.

3. **Use Multiple Indicators for Fraud Assessment**

Transaction amount and account balance did not clearly distinguish fraudulent from non-fraudulent transactions in the descriptive analysis. Therefore, fraud assessment should not rely on a single characteristic. A combination of risk score, failed transaction activity, transaction details, and other relevant indicators should be evaluated to support more comprehensive assessments.

4. **Investigate Unusual or Incomplete Category Information**

The Unknown card type category recorded a fraud rate of 43.33%, but it contained only 30 transactions. Rather than concluding that this category presents greater fraud risk, further investigation should determine why the card type was unavailable or unclassified. Improving data completeness and monitoring unusual category values could support more reliable analysis.

5. **Review Fraud Patterns Across Categories**

Fraud rates were relatively similar across locations, transaction types, card age groups, and authentication methods. These characteristics should continue to be monitored, but the observed differences should not be treated as strong indicators of fraud on their own. Further analysis could examine combinations of variables to identify patterns that may not be visible in individual category comparisons.

6. **Establish Ongoing Monitoring of Fraud Trends**

Monthly transaction volumes and fraud rates fluctuated during 2023, although no consistent upward or downward trend was observed. Regular monitoring of fraud rates, transaction volumes, and changes in risk indicators could help identify emerging patterns and support timely investigation.

7. **Validate Future Fraud Detection Models**

The current project identifies associations through data cleaning, exploratory analysis, and visualisation. As a next step, supervised machine learning models could be developed and evaluated using appropriate training and testing datasets. Model performance should be assessed using metrics such as precision, recall, F1-score, and the precision-recall curve, alongside a review of false positives and false negatives. This would help determine whether the identified variables can support reliable fraud detection in unseen transactions.

8. **Maintain Data Quality and Integrity**

The data cleaning process removed missing values through appropriate imputation, checked for duplicate records, validated binary and numerical fields, and standardised timestamps and categorical values. These data quality checks should be maintained in future analyses because incomplete, inconsistent, or invalid data can affect the reliability of fraud-related findings.

In summary, the recommendations focus on prioritising transactions for review, examining multiple fraud indicators, maintaining data quality, monitoring changes over time, and validating future detection models. These steps provide a foundation for further investigation into how the observed patterns could be used in a practical fraud detection approach.

13. Conclusion

This project focused on analysing transaction data to identify patterns and characteristics associated with fraudulent activity. Using a dataset containing 50,000 transactions and 21 columns, the project involved data cleaning, preprocessing, exploratory data analysis, and visualisation using Python in Google Colab.

The data cleaning process addressed missing values, checked for duplicate records, validated numerical and categorical fields, and ensured that timestamps and fraud labels were suitable for analysis. Exploratory data analysis and visualisations were then used to examine transaction distributions, compare fraud rates across categories, investigate monthly trends, and explore relationships between numerical variables.

The analysis found that 16,067 transactions (32.13%) were labelled as fraudulent. Risk score and failed transaction count over the previous seven days showed the strongest positive correlations with fraud status among the numerical variables examined, at 0.39 and 0.51, respectively. Transactions in the High Risk category also had a higher observed fraud rate of 56.02%, compared with approximately 20% in the Low and Medium categories. In contrast, fraud rates across locations, transaction types, authentication methods, card age groups, and recognised card types were relatively similar. Monthly fraud rates fluctuated throughout 2023 without showing a consistent upward or downward trend.

Overall, the project demonstrates how data cleaning, exploratory analysis, and visualisation can help reveal patterns in transaction data and support a better understanding of fraudulent activity. However, the findings describe associations within the available dataset and do not establish causation or demonstrate the performance of a predictive fraud detection model.

Future work could build on these findings by developing and evaluating machine learning models, testing their performance on unseen data, and examining how multiple transaction characteristics can be combined to identify potentially fraudulent transactions more reliably. This would provide a further step towards assessing the practical value of data-driven fraud detection approaches.

14. Limitations and Future Scope

14.1 Limitations

Although the project identified several patterns associated with fraudulent transactions, certain limitations should be considered when interpreting the findings.

- **Dataset Scope:** The analysis was based on a dataset containing 50,000 transactions. The findings may not represent fraud patterns across all financial institutions, customer groups, or transaction environments.
- **Limited Predictive Analysis:** The project focused on data cleaning, exploratory data analysis, and visualisation. A machine learning model was not developed and evaluated; therefore, the analysis does not establish the accuracy or reliability of a fraud detection system.
- **Association Does Not Establish Causation:** Relationships observed between fraud status, risk score, and failed transaction counts indicate associations within the dataset. They do not establish that these variables cause fraudulent activity.
- **Limited Category Differences:** Fraud rates were relatively similar across several categories, including location, transaction type, and authentication method. These descriptive comparisons may not reveal more complex relationships between combinations of variables.
- **Small Sample in the Unknown Card Category:** The Unknown card type category contained only 30 transactions. Its observed fraud rate of 43.33% may not be representative because of the small number of records.
- **Dependence on Data Quality:** Although missing values, duplicates, and invalid values were checked during preprocessing, the analysis remains dependent on the accuracy and completeness of the original dataset. Any limitations in the source data may affect the results.

14.2 Future Scope

The project can be extended through additional analysis and the development of more advanced fraud detection techniques.

- **Machine Learning Model Development:** Classification algorithms such as Logistic Regression, Decision Trees, Random Forest, and Gradient Boosting could be explored to classify transactions as fraudulent or non-fraudulent.
- **Model Evaluation:** Future work could evaluate models using precision, recall, F1-score, confusion matrices, and precision-recall curves. These metrics would help assess fraud detection performance and the balance between missed fraud cases and false alarms.
- **Feature Engineering:** Additional features could be developed from existing transaction information, such as transaction frequency patterns, combinations of failed transactions and risk scores, or time-based behaviour, where supported by the available data.

- **Advanced Pattern Analysis:** Further analysis could investigate interactions between multiple variables to determine whether combinations of transaction characteristics provide more useful insights than individual variables considered separately.
- **Larger and More Diverse Datasets:** Testing with additional datasets from different periods, transaction environments, or financial institutions could help assess whether the observed patterns remain consistent across different settings.
- **Ongoing Monitoring and Visualisation:** The analysis could be extended into an interactive dashboard to monitor transaction volumes, fraud rates, risk categories, and changes in fraud-related indicators over time.
- **Model Explainability:** If a predictive model is developed, explainability techniques could be used to understand which variables contribute most to its predictions and help make its outputs easier to interpret.

In conclusion, the current project provides a foundation for understanding transaction data and identifying characteristics associated with fraud. Future work involving predictive modelling, broader datasets, and more advanced analysis could help determine whether these findings can support a reliable and practical fraud detection approach.

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